Media 100

User Guide

A Business Unit of BorisFX

BORIS FX SOFTWARE

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Media 100
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2

Project Settings and Preferences

Introduction

The first step in creating video and audio compositions in Media 100™ is to create a *project*. A project controls, tracks, and displays all the elements you assemble for a video composition.

Within a project, you create *bins* and *programs*.

- A bin contains video and audio clips and other media.
- A program is where you arrange and edit clips on a timeline.

This chapter explains how to create a Media 100 project. It helps you

- Plan a project
- Create a new project
- Prepare a project for importing media and editing
- Set up the Quick Launch feature

Planning a Project

Before beginning a project, determine what input sources and output formats you plan to use to ensure the best final output.

In the preproduction phase of your project it is important to shoot and prepare your source footage with editing in mind. While planning your project, consider the issues in this section.

Careful planning before you begin creating a project helps eliminate the need to reacquire media part way through the project.

NOTE Existing video, photographic, and audio materials are generally protected under copyright law. Secure permission from the copyright owners before incorporating any such material into a program.

Source Video

Media 100 supports both digital and analog source in 8-bit and 10-bit uncompressed and compressed native HD and SD video signals. The system also integrates full support for offline formats, allowing the capture of HD content as SD offline-quality media. The SD offline programs can then be automatically conformed at full uncompressed HD resolution.

Media 100 also supports real-time broadcast format conversions between HD and SD formats. This conversion is available upon input as well as output. There are independently assignable digital and analog output signals, all simultaneously active. Video Input Settings are modified in the Video Input panel of the Media 100 Preferences.

Video Source	Source Standard	Media 100 SDe Media Standard (Conversion Options)	Media 100 HDe Media Standard (Conversion Options)	Media 100 HD Suite Media Standard (Conversion Options)
SD Composite NTSC or PAL 4:3 or 16:9	■ SD 720x486i (NTSC) SD 720x576i (PAL) ■ HD 1920x1080i	■ SD 720x486i ■ SD 720x576i 16x9 or 4x3	■ SD 720x486i ■ SD 720x576i 16x9 or 4x3	■ NA
SD Component NTSC or PAL 4:3 or 16:9	■ SD 720x486i (NTSC) SD 720x576i (PAL) ■ HD 1920x1080i	■ SD 720x486i ■ SD 720x576i 16x9 or 4x3	■ SD 720x486i ■ SD 720x576i 16x9 or 4x3	■ NA
SD Serial Digital NTSC or PAL 4:3 or 16:9	■ SD 720x486i (NTSC) SD 720x576i (PAL) ■ HD 1920x1080i	■ SD 720x486i ■ SD 720x576i 16x9 or 4x3	■ SD 720x486i ■ SD 720x576i 16x9 or 4x3	■ HD 1920x1080i ■ SD 720x486i ■ SD 720x576i 16x9 or 4x3
SD Y/C	■ SD 720x486i (NTSC) SD 720x576i (PAL) ■ HD 1920x1080i	■ SD 720x486i ■ SD 720x576i 16x9 or 4x3	■ SD 720x486i ■ SD 720x576i 16x9 or 4x3	■ NA
SD FireWire	■ SD 720x486i (NTSC) SD 720x576i (PAL) ■ HD 1920x1080i	■ SD 720x486i ■ SD 720x576i 16x9 or 4x3	■ SD 720x486i ■ SD 720x576i 16x9 or 4x3	■ SD 720x486i ■ SD 720x576i 16x9 or 4x3
HD Serial Digital Component	■ HD 1920X1080i 16x9 ■ HD 1280X720p 16x9	■ NA	■ HD 1920X1080i 16x9 ■ SD 720x486i ■ SD 720x576i 16x9 or 4x3	■ HD 1920X1080i 16x9 ■ SD 720x486i ■ SD 720x576i 16x9 or 4x3
HD Serial Digital	■ HD 1920X1080i 16x9 ■ HD 1280X720p 16x9	■ NA	■ HD 1920X1080i 16x9 ■ SD 720X486 16x9 or 4x3	■ HD 1920X1080i 16x9 ■ SD 720X486 16x9 or 4x3

NOTE Media 100 systems allow for one conversion. Therefore, if you convert the source content to a different standard during acquisition or import, that is the standard the content must be output to. However, if the source content is acquired in it's native form it can be converted on output. Again, only one conversion is allowed. (Media 100 HDx enables multiple conversions.) Also note, the conversion options that are available are dependent upon the Media 100 system being used.

NOTE When converting between 4:3 and 16:9 aspect ratios, additional conversion options are enabled. These include 4:3 to 14:9, 16:9 to 14:9, Crop, Pillarbox and Scale to Fit. In addition a Custom Scaler offers complete flexibility of conversion options.

TIP All clips in a program must use one video standard. If a clip to be used in a program is a different standard, Media 100 offers the ability to conform the media to the program standard. When a clip with one standard is dropped in a timeline with another standard, a Conform dialog box opens. This enables the media to be conformed by scaling the clip or by reacquiring the content.

Source Audio

Media 100 supports AES/EBU digital, balanced analog source audio and FireWire. When selecting audio files for your project, consider the following:

- Decide the audio sample rate to use for the entire project. You can not use audio files that are sampled at different rates in the same program.

  The Media 100 system will resample audio upon acquisition to your desired sample rate (11025 Hz, 22050 Hz, 32000 Hz, 44100 Hz, 48000 Hz).

- If the source audio uses different sample rates, determine which files, if any, you can re-sample to achieve one rate for the project.

Audio Input Settings are modified in the Audio Input panel of the Media 100 Preferences (Media 100>Preferences>Audio Input).

NOTE Although you may convert the audio sample rate on acquisition, all audio is handled at 48 kHz through the Media 100 hardware.

NOTE Media 100 HD supports SDI embedded audio.

Computer Graphics

Keep in mind video and computer signals are fundamentally different. Video signals are comprised of non-square pixels, while computer-based pixels are square. If you plan to use many square-pixel still images created in a third-party product, such as Adobe® Photoshop®, determine the most effective approach for their use.

- Design elements in these applications for square pixel resolution and enable Media 100 to apply the necessary scaling on import to convert the aspect ratio.
- Design elements at a resolution of 648x486 for NTSC SD 4:3. This will eliminate the possibility of jaggies as the interpolation takes place along with the scanline, not against it. The mapping of pixels then from 648 to 720 is just a small scale of the pixel aspect ratio.
- Another popular resolution is 720x540 for NTSC SD 4:3, however there is a bit more room for error on the scaling down to 486.

TIP If you are working on an element with horizontal lines, stretch the resolution from 648 to 720 so you are working at 720x486. This will aid in the elimination of buzzing.

Video Standard	Square Pixel Graphic Resolution
NTSC SD 4:3	648 x 486
NTSC SD 16:9	864 x 486
PAL SD 4:3	768 x 576
PAL SD 16:9	1024 x 576
HD 1920 x 1080	1920 x 1080
HD 1280 x 720	1280 x 720

NOTE HD resolutions are already square pixel so work in the native resolution. No scaling is required.

Output Formats

Media 100 enables simultaneous output of digital signals, analog signals and a reference out. You may elect to output SD, HD 720p or HD 1080i. Keep in mind for output you are allowed only one conversion of format. Therefore if you perform a conversion on acquistion, for example an SD signal to HD 1080i that is using your one conversion of format and thus you could not cross-convert on output to 720p. Output Settings are modified in the Audio Output and Video Output panels of the Media 100 Preferences (Media 100>Preferences).

Video Output	Video Output Options
Output	■ Media 100 system ■ FireWire ■ FireWire (Mastering Only) ■ None
SDI	■ SD 720x486i (NTSC); SD 720x576i (PAL) ■ HD 1920x1080i ■ HD 1280x720p
Analog	■ SD Composite ■ SD Component ■ SD Y/C ■ HD Component

NOTE When converting from 4:3 SD media to HD output, additional conversion options are enabled. These include 4:3 to 14:9, 16:9 to 14:9, Crop, Pillarbox and Scale to Fit. When converting HD media to SD output, additional conversion options of 4:3 to 14:9, 16:9 to 14:9, Crop to 4:3, Letterbox to 4:3 and Scale to Fit 16:9 are also available. In addition a Custom Scaler offers complete flexibility of conversion options.

NOTE If FireWire is the output selection and a deck is not connected, the timeline will not play because it’s attempting to master the timeline as output and it can’t. Therefore, if FireWire (Mastering Only) is the selected output, the system will allow the timeline to be played and will seek the deck connection when mastering only.

▼ About the Interface

The Media 100 interface complements the Mac® OS X™ operating system. Some of the OS X features that Media 100 has implemented are:

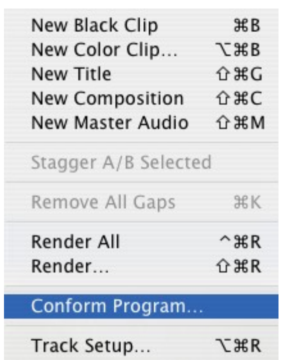
- Flashing buttons that require action
- Window tabs
- Two-button mouse contextual menus

TIP For a one button mouse, click and press the HELP key to access contextual menus.

- Mouse buttons mapped to the application

Using Contextual Menus

Media 100 provides contextual menus in the Program, Bin, and Project windows. To access the contextual menus, press HELP and click the mouse in the Program, Bin or Project window. The following example is a Program window contextual menu.



Creating a Project

To start a new project, create, name, and save three elements: the project file, one or more bins, and one or more programs. As you create these elements, Media 100 generates a separate window for each.

Project window	Maintains a list of all the bins and programs in a project.
Bin window	Contains acquired clips or other media that you store in the bin. You can create an unlimited number of bins.
Program window	Displays a representation of the clips you assemble and manipulate. You can create an unlimited number of programs.

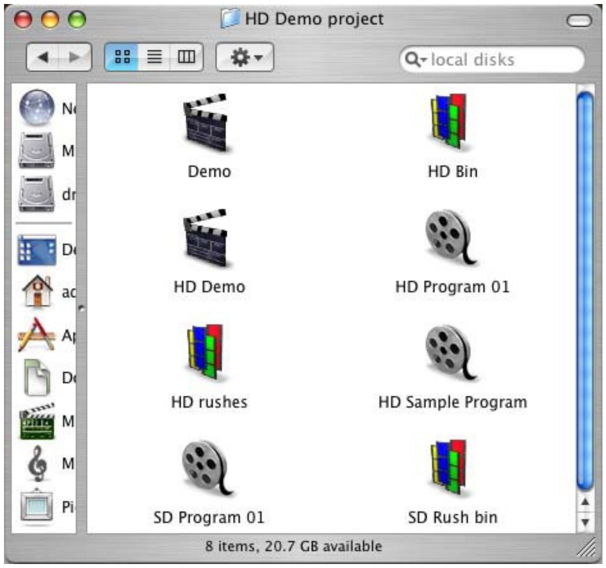
When you create each element, Media 100 stores them as files on your system. To locate these files easily, create a separate folder for each of your projects.

NOTE You can use the same bins and programs in multiple projects. The media folders are shared.

Setting Up Project Folders

To set up a project folder, decide where you want to store your project. For example, to create a project called "Demo," first create a folder called "Media 100 Projects." Then, within this folder, create another folder called "Demo."

Store your Demo project, bin, and program files in the Demo folder. Media 100 automatically stores these files in Icon view.

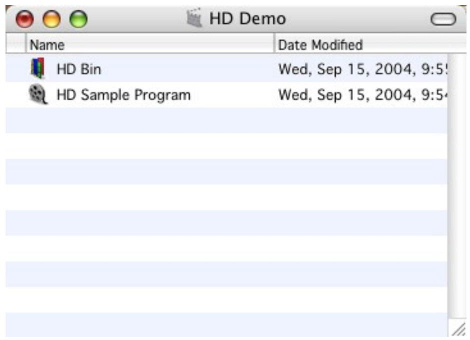


Memory Requirements

The Media 100 application requires 2GB (Gigabyte) of Random Access Memory (RAM). If you do not have at least 2GB of RAM, system performance may be affected.

Creating a New Project

The Project window is the primary interface to the project and remains open while you work with its bins and programs. As you create bins and programs for your project, the Project window displays a list of all the associated bins and programs.



You cannot open more than one project at a time. When you close a project, you also close associated bins and programs. You cannot open a bin or program without having a project open.

To create a new project for the first time

1 Launch Media 100.

The startup screen appears, and a directory dialog box opens.

2 Type a name for your project, select a location to store it, and click Save.

The Project Settings dialog box appears.

NOTE Unless you plan to use another system, save the project files to your internal drive. This makes it easier to troubleshoot media problems, and may improve system performance.

Relaunching Media 100

Press the SHIFT key while launching Media 100 to open the application without opening the last saved project files (bins and programs).

Press the OPTION key while launching Media 100 to open the application without opening the last saved project. The Open window appears.

Choosing Project Settings

The Project Settings dialog box has four panels.

Media Destinations	Defines where on your system to store media files.
Media Locations	Lists the location of your stored media files.
Media Settings	Defines compression settings that control the quality of acquired or imported video material, as well as rendered media such as transition effects, and graphics.
Real Time	Plays real-time effects and titles at Draft Quality

To open the Project Settings dialog box

1 Press ⌘-M or choose Media 100>Project Settings.

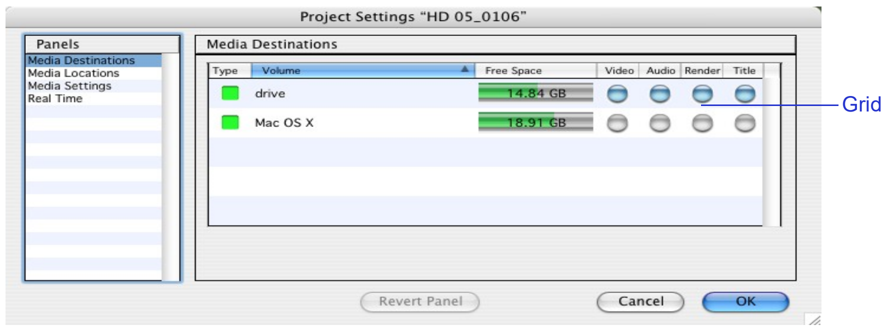
2 Click a panel name in the Panels list to switch to that panel.

Media Destinations

Specify where to store your media in the Media Destinations panel. Where you store media affects the speed with which Media 100 accesses the media. Store all content on high-speed external drives.

To specify Media Destinations

➤ In the Media Destinations panel, click the intersecting lines in the grid to select the external drive(s) where you want to store the following types of media files: Video, Audio, Render, and Title. Render is where rendered effects, filters and all other rendered clip media will reside. Title is where where titles created using the Boris tools will reside. Render and Title media will be placed in the project media folder in a subfolder labeled "Rendered Media" and Titles" respectively.

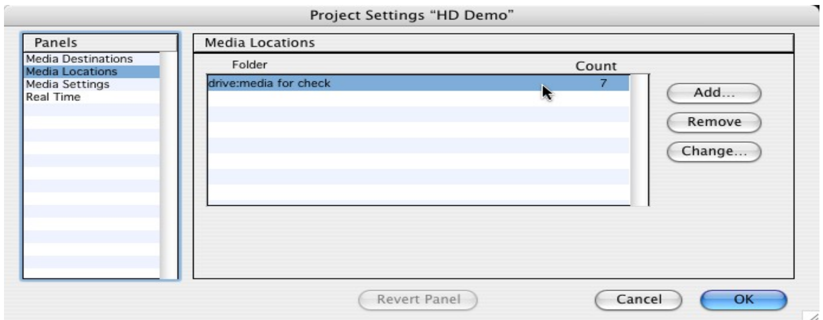


Teal spheres appear where you click the grid, indicating the selected drives for the acquired media files.

If you do not specify where to store media file types for a new project, the system defaults to the disk where the new project resides, which may be the internal system drive.

Media Locations

The Media Locations panel lists where your media is located. When you use the Find Media command, Media 100 searches the folders in the Media Locations panel.



NOTE An italicized folder name means that the system could not find the folder.

Media Settings

The Media Settings panel is where you choose a compressor to be used for acquired or imported media as well as a compressor for rendered media. Settings are specified separately for SD media and HD media.

To choose a compression type

➤ Choose a compressor type for acquisition, importing and FireWire output from the Compression Type menu. Choose a compressor type for rendered media from the Compression Type menu. These settings controls how the Media 100 system compresses your video data for acquisition, import, FireWire output and rendered media respectively.

Choose...	To...
Media 100 HD™	Create the best quality content **NOTE** This is the only available compression when acquiring HD content
Media 100 i™	Take your final output to a Media 100 i system
Media 100 844/X™	Take your final output to a Media 100 844/X system.
DV	Supports SD only; Good Quality; 3.13 MB/sec data rate
DVCPro	Supports SD only; Good Quality
DVCPro 50	Supports SD only; Very Good Quality; 6.26 MB/sec data rate
DVCPro HD	Suports HD only; Excellent quality; 12 MB/sec
Motion JPEG B	Supports SD or HD; Very Good Quality
Uncompressed 8-bit 4:2:2	Supports SD or HD; Excellent quality; 21 MB/sec data rate for SD; 100-124 MB/sec data rate for HD
Uncompressed 10-bit 4:2:2	Supports SD or HD; Excellent quality; 28 MB/sec data rate for SD; 133-166 MB/sec data rate for HD

NOTE Compression Type options vary depending on the video standard selected (Media 100>Preferences>Video Input). For example if HD 1920x1080i is the selected standard, the compression type options are: Media 100 HD; Motion JPEG B; Uncompressed.

Codec Selections by Model

CODEC	SDe SD	SDe HD	HDe SD	HDe HD	HDSuite SD	HD Suite HD
Media 100 HD™	X	X	X	X	X	X
Media 100 i™	X		X		X	
Media 100 844/X™	X		X		X	
DV	X		X		X	
DVCPro	X		X		X	
DVCPro 50	X		X		X	
DVCPro HD		X				X
Motion JPEG B	X	X	X	X	X	X
Uncompressed 8-bit 4:2:2	X	X	X	X	X	X
Uncompressed 10-bit 4:2:2	X	X			X	X

Setting the Quality Data Rate

The Quality settings vary based on the compressor selection. The Media 100 HD compressor is the native compressor for the Media 100 system. When choosing the Media 100 HD compressor, the quality data rate of the acquired content can be either 8-bit or 10-bit uncompressed. You may also elect to acquire at draft quality which field doubles the media on acquisition (acquiring a single field and doubling it). The 844/X compressor enables Draft Quality, 8-bit uncompressed or 10-bit uncompressed quality options. The Media 100 i compressor enables Draft Quality and compression options from 10 KB per frame to 300 KB per frame as well as Lossless Compression. The DV, DVCPro, DVCPro 50 and Motion JPEG B compressors enable a slider option from Least to Best.

The target data rate (kilobytes per frame) directly affects the amount of disk space each frame of the source media occupies. The target data rate also determines the quality of the image produced when the media is acquired and compressed.

Lower target data rates let you store more data on your disk at a lower image quality. As your data rates rise, the image quality improves, and the disk storage capacity decreases.

8-bit versus 10-bit

One bit of data can define two levels, for example black and white. Two bits of data can define four levels, three bits of data define eight levels and so on. Therefore, eight bits can define two hundred fifty six levels and ten bits can define one thousand twenty four levels.

8-bit images use 8 bits of data per channel which delivers a data range between 0 and 255 for each channel. 10-bit images use 10 bits of data per channel which delivers a data range between 0 and 1023 for each channel. Therefore a 10-bit image can carry four times more data per channel than an 8-bit image.

Every video signal is comprised of luminance and chrominance. Luminance is the brightness of the signal from black to white and chrominance is the color part of the signal in regards to hue and saturation. There is one luminance (Y) part and two chrominance (Cr, Cb) parts to every video signal. Every line of a standard definition video image has 720 luminance samples and 360 chrominance samples each for Cr and Cb. This totals 1440 samples per line.

In a NTSC 525/60 (720 x 486) picture there are 486 active lines. Therefore, 486 active lines sampled at 1440 per line means there are 699,840 pixels per picture. If each pixel is sampled at 8-bit resolution, the total number of bits is 5,598,720 or 699,840 bytes or 699.8 KB (kilobytes) per frame. At 29.97 frames per second, the total throughput is 20,973 KB or 21 MB (megabytes) per second. At 10-bit resolution, there are still 699,840 pixels per picture, however sampled at 10-bit resolution, the total number of bits is 6,998,400 or 874,800 bytes or 874.8 KB per frame. At 29.97 frames per second, the total throughput is 26,224 KB or 26 MB per second.

The same can be applied to a PAL 625/50 (720 x 576) picture which is comprised of 576 active lines. Sampled at 1440 per line there are 829,440 pixels per picture. At 8-bit resolution, the total number of bits is 6,635,520 bits or 830 KB. At 25 frames per second, the total throughput is 20,750 KB or 21 MB per second. At 10-bit resolution, the total number of bits is 8,294,400 or 1,036,800 bytes or 1036.8 KB per frame. At 25 frames per second, the total throughput is 25,920 KB or 26 MB per second.

As 8-bit quality for both NTSC and PAL is approximately 21 MB per second which is equivalent to 1,260 MB per minute or 75,600 MB per hour. This means one hour of 8-bit video content will require approximately 76 GB of drive space.

As 10-bit quality for both NTSC and PAL is approximately 26 MB per second which is equivalent to 1560 MB per minute or 93600 MB per hour. This means one hour of 10-bit video content will require approximately 94 GB of drive space.

In an HD 1920 X 1080 picture there are 1080 active lines. Therefore, 1080 active lines sampled at 1920 per line means there are 2,073,600 pixels per picture. If each pixel is sampled at 8-bit resolution, the total number of bits is 16,588,800 or 2,073,600 bytes or 2073.6 KB (kilobytes) per frame. At 29.97 frames per second, the total throughput is 62,145.8 KB or 62 MB (megabytes) per second. At 10-bit resolution, there are still 2,073,600 pixels per picture, however sampled at 10-bit resolution, the total number of bits is 20,736,000 or 2,592,000 bytes or 2,592 KB per frame. At 29.97 frames per second, the total throughput is 77,682 KB or 78 MB per second.

The throughput at 8-bit quality for HD 1920 x 1080 is approximately 62 MB per second which is equivalent to 3720 MB per minute or 223,200 MB per hour. This means one hour of HD 1920 x 1080 video content will require approximately 223 GB of drive space.

NOTE The data rate applies only to video source material. It does not apply to audio material, as acquired audio files are not compressed.

Using Draft Modes

The batch acquire feature gives you the flexibility to initially acquire your video media at a draft-quality setting and then reacquire at the online-quality setting.

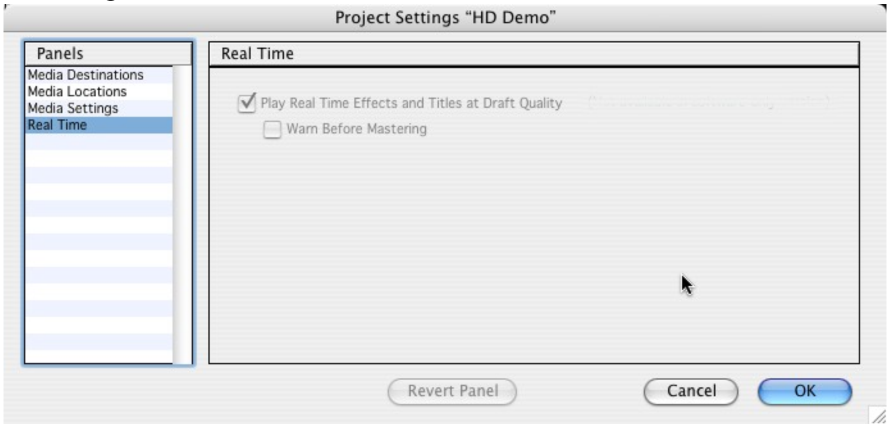
Video media acquired using draft-quality settings results in lower picture quality and lack of detail unsuitable for final output. Draft quality settings use field doubling at playback as only a single field (the dominant field) is acquired. Draft quality settings also use higher compression ratios when the video source is acquired using a compressor that enables compression such as Media 100 i or Motion JPEG B.

Therefore, by using draft quality, more content may be stored on the disk arrays for the editing process. But keep in mind the content will need to be reacquired at an uncompressed or higher quality compressed rate for final delivery.

NOTE You cannot use batch acquire for audio source media without integrated timecode.

Real Time

The Real Time panel is where you specify the playback of real-time effects and titles to occur at draft quality. When a compressor other than the native Media 100 HD format is used, this enables smoother playback as each frame is decompressed on the fly during the playback process. If you select this option, you can also choose to Warn Before Mastering as a reminder to deselect the draft quality playback when mastering.



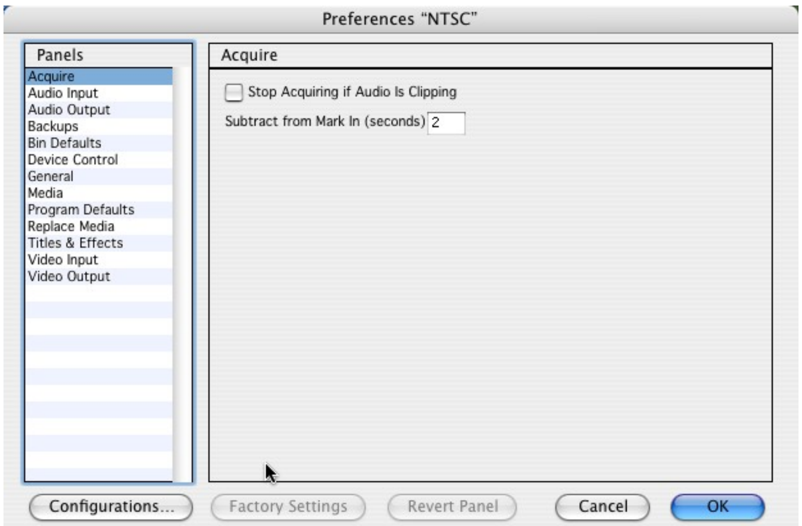
Setting Preferences

Before creating bins or programs for your project, configure your Media 100 system by selecting settings from the Preferences dialog box.

To access the Preferences dialog box

➤ Choose Media 100>Preferences.

The Preferences dialog box appears.



The following table explains elements common to all panels.

Preferences Dialog Box Elements

Name	Description
Panels list	Displays the Preferences panel names. Click a panel name to switch to that panel.
Configurations button	Accesses the Configurations dialog box to use customized preference settings.

Preferences Dialog Box Elements *(Continued)*

Name	Description
Factory Settings button	Changes the settings in the current panel to the Media 100 HD default settings.
Revert Panel button	Changes the settings in the current panel to the previously saved settings.
Cancel button	Closes the dialog box without saving changes to any panel.
OK button	Saves the settings in all panels to the active configuration and closes the dialog box.

Working with Preference Configurations

Media 100 provides two preference configurations: NTSC and PAL. You can customize and save preference configurations. Each configuration contains a particular set of Preference settings. Only one preference configuration is active at a time. The active configuration remains in effect until you change or delete it.

You can set custom configurations to use for different types of projects. This feature is also useful if you are working in a multi-user environment, where each user has individual project requirements.

NOTE If you are working in a multi-user environment or on multiple projects, when you open a project, verify that the correct configuration is active.

You can

- Create a custom configuration
- Set a preference configuration
- Export and import a configuration
- Rename a configuration
- Delete a configuration

Preference configurations are located in the Media 100 Preferences folder. The Media 100 Preferences folder is located in the User Account/Library/Preferences/Media 100 Preferences.

Creating a Custom Configuration

Create a configuration using one of the following methods:

- Change and rename

  Change the settings of the current configuration, then rename your new settings.

- Duplicate and change

  Duplicate a configuration, rename it, then change the settings as needed.

To create a configuration by changing and renaming

1. Choose Media 100>Preferences.

   The Preferences dialog box appears.

2. Make the necessary changes to the current configuration.

3. Click Configurations.

   The Confirm dialog box appears.

4. Type a new name in the name field.

   The Save button changes to a Create button.

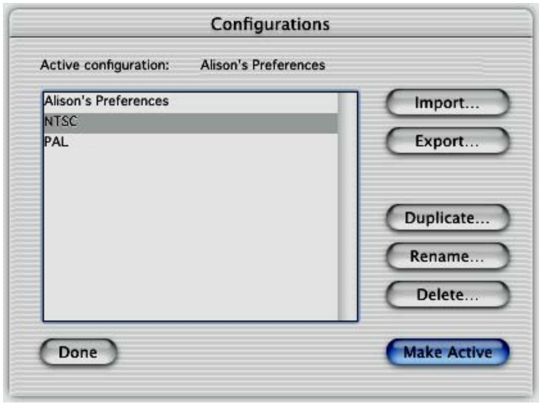
5. Click Create.

   The new configuration appears in the Configurations list.

To create a configuration by duplicating and changing

1 Choose Media 100>Configurations>Edit Configurations.

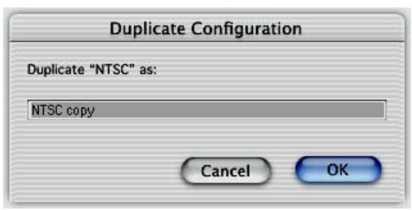
The Configurations dialog box appears.



2 Select a configuration from the Configurations list.

3 Click Duplicate.

The Duplicate Configuration dialog box appears.



4 Type a new name in the name field and click OK.

Media 100 returns you to the Configurations dialog box.

NOTE If you type a name that has illegal characters or is already in use, an error message appears. Click OK and type another name.

5 Click Make Active.

6 Make the necessary changes in the Preferences dialog box and click OK.

Setting a Preference Configuration

Media 100 provides two configurations: NTSC and PAL. You can use one of these, or you can create or select a custom configuration.

To set a preference configuration

➤ Choose Media 100>Configurations and select a configuration from the submenu.

Or

1 Open the Configurations dialog box by doing one of the following:
   - Choose Media 100>Configurations>Edit Configurations.
   - Choose Media 100>Preferences and click Configurations.

2 Select a configuration from the list.

3 Click Make Active.

The selected configuration becomes active.

TIP Although you can change the NTSC and PAL configurations, you may want to create a custom configuration when you alter the settings to maintain the integrity of the original.

Exporting and Importing Configurations

Export a configuration to a disk or shared volume for use on another Media 100 system or as a method of backup.

To export a configuration

1. In the Configurations dialog box, select a configuration and click Export.
2. Select a destination in the Export dialog box and click Export.

Once you export a configuration, you can import it.

To import a configuration

1. Choose Media 100>Configurations>Edit Configurations.

   The Configurations dialog box appears.

2. Click Import.
3. Select the configuration location in the Import dialog box and click Import.

Renaming a Configuration

You can rename a configuration to make the name more relevant.

To rename a configuration

1. In the Configurations dialog box, select the configuration and click Rename.
2. Change the name in the Rename dialog box and click OK.

NOTE If you type a name that has illegal characters or is already in use, an error message appears. Click OK and type another name.

Deleting a Configuration

Delete a configuration when it is no longer useful.

TIP To delete a configuration from the current list, but save it for future projects, export the configuration to disk, then delete it from the Configurations list.

To delete a configuration

1 In the Configurations dialog box, select the configuration and click Delete.

2 In the Confirm dialog box that appears, click Delete.

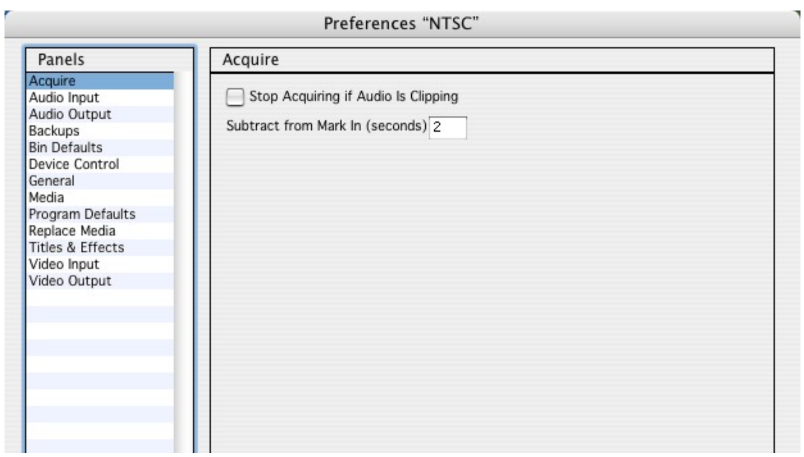
NOTE Although you can delete the NTSC and PAL configurations, it is not recommended, as you may need them in the future. To restore them, you need to trash your Media 100 Preferences folder (User Account/Library/Preferences/Media 100 Preferences). You will lose any custom configurations that you do not move or export elsewhere.

Preference Panels

The following sections explain the preference options in each panel.

Acquire Panel

The Acquire panel contains settings for acquiring media.



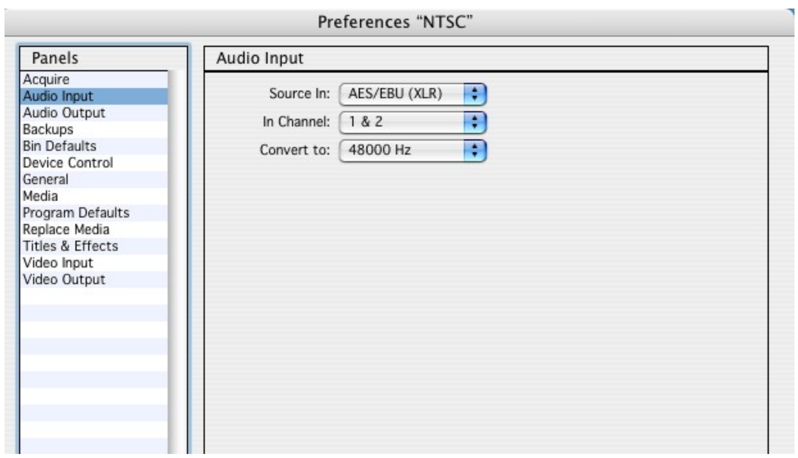
The following table describes the Acquire panel elements.

Acquire Panel Elements

Name	Description
Stop Acquiring if Audio Clips check box	Stops the acquisition process when audio is clipping until you press Start in the Edit Suite window.
Subtract from Mark In (secs) field	Sets the time (from 1 to 5 seconds) subtracted from the In mark in the timecode field when using the Mark In button to set a start point for acquiring media. This compensates for the delay between when you see the desired start point and when you click Mark In.

Audio Input Panel

The Audio Input panel contains settings for acquiring audio media.



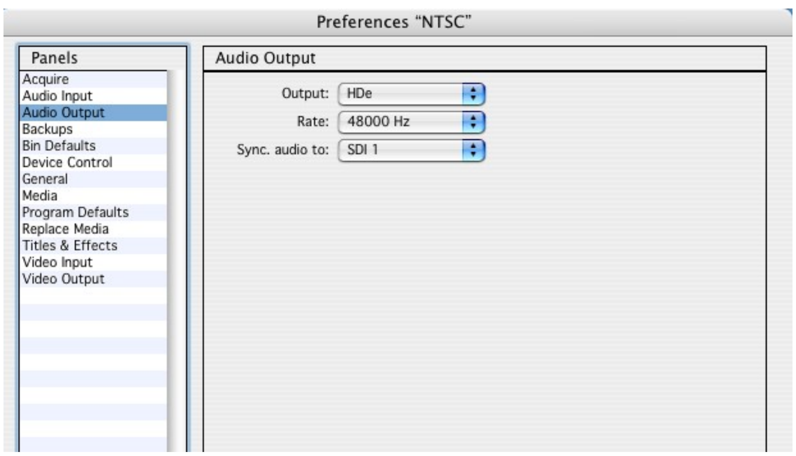
The following table describes the Audio Input panel elements.

Audio Input Panel Elements

Name	Description
Source In menu	Specify the audio input source: Analog; AES/EBU via XLR connectors; AES/EBU via BNC connectors; SDI Embedded*; FireWire; None **NOTE** *SDI Embedded is available on Media 100 HDx.
In Channel menu	Specify the channels through which the audio source is being acquired.
Convert to Sample Rate menu	Specify the desired sample rate to convert the audio clip. **NOTE** This setting determines the audio sample rate when creating a new program timeline as well.

Audio Output Panel

The Audio Output panel contains settings for audio output.



The following table describes the Audio Output panel elements.

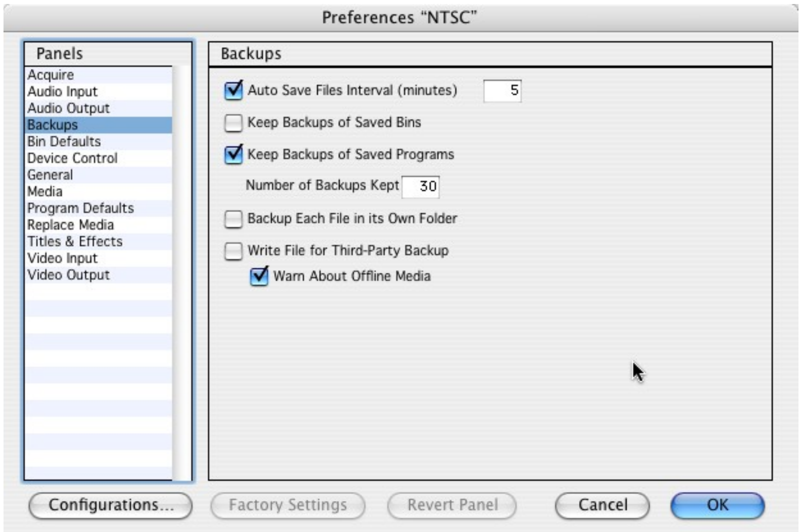
Audio Output Panel Elements

Name	Description
Output	Specifies the Media100 system; FireWire; FireWire (Mastering Only) or None for audio output.
Sample Rate	All audio output through the Media 100 hardware is sampled at 48 kHz.
Sync audio to	Specify the video output to sync the audio to.

NOTE If FireWire is the output selection and a deck is not connected, the timeline will not play because it’s attempting to master the timeline as output and it can’t. Therefore, if FireWire (Mastering Only) is the selected output, the system will allow the timeline to be played and will seek the deck connection when mastering only.

Backups Panel

The Backups panel contains project backup and auto save settings. The actual folder in which the backup selections are stored is located in User Account/Documents/ Media 100 Backups.



The following table describes the Backups panel elements.

Backups Panel Elements

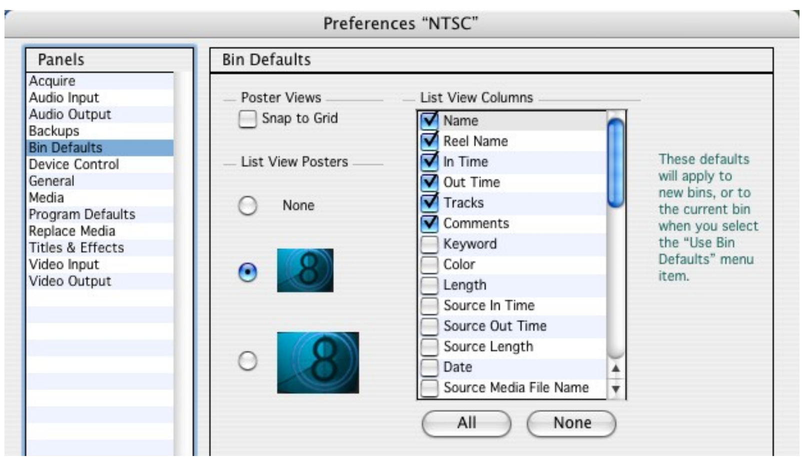
Name	Description
Auto Save Files Interval check box and field	Saves all your project files at the interval you type in the field. Specify a value from 5 to 60 minutes. Media 100 will activate auto save only after you save a new file at least once.
Keep Backups of Saved Bins check box	Creates a backup copy of a bin when you save it (or when autosave activates) and stores it in the Backups folder in the Media 100 folder.
Keep Backups of Saved Programs check box	Creates a backup copy of a program when you save it (or when autosave activates) and stores it in the Backups folder in the Media 100 folder.

Backups Panel Elements *(Continued)*

Name	Description
Number of Backups Kept field	Limits the combined number of bin and program backup copies to the specified number (from 1 to 99). The Backups folder stores the backup files by date modified, with the most recent file listed first. Once the limit is reached, the system replaces the oldest files first.
Backup Each File in its Own Folder check box	Creates a folder under the Backups folder for each program with the name of the program. Once the quantity of backups specified in Number of Backups Kept is reached, the oldest version of the program in that folder is deleted.
Write File for Third-Party Backup check box	Creates a database of project information.
Warn About Offline Media check box	Alerts you that you have offline media when backing up a project.

Bin Defaults Panel

The Bin Defaults panel contains default settings for new bins.



The following table describes the Bin Defaults panel elements.

Bin Defaults Panel Elements

Name	Description
Snap to Grid check box	In Poster view, confines movement of clips to a grid.
List View Posters	In List view, determines the size of the poster, if any.
List View Columns	The check boxes selected in this section represent the columns of clip information displayed and their order when bins are in list view.
Name	Displays the clip name.
Reel Name	Displays the name of the reel from which the media was acquired.
In Time	Displays the In point time of a clip.
Out Time	Displays the Out point time of a clip.
Tracks	Displays the audio and video track(s) to which the media is assigned.
Comments	Displays comments that you typed into the field providing clarifying or instructive information about the clip.
Media Online	Allows you to categorize all the online media in your project. Any media not listed as online is offline. This media you need to relink or reaquire.
Frame Size	Displays the frame size of a clip.
Aspect Ratio	Displays the aspect ratio of a clip.
Data Rate	Displays the data rate - uncompressed or compressed with the KB rate and mode (Draft or Online) at which the clip was acquired. Or, in the case of logged clips, the KB rate at which the clips will later be acquired.
Keyword	Displays a key word that is a representative word for the clip.
Color	Displays a selectable color tag for each clip.

Bin Defaults Panel Elements *(Continued)*

Name	Description
Length	Displays the length of a clip.
Source In Time	Displays the In time of the source media, regardless of trimming.
Source Out Time	Displays the Out time of the source media, regardless of trimming.
Source Length	Displays the length of the source media, regardless of trimming.
Date	Displays the date the clip was acquired or rendered.
Source Media File Name	Displays the file name of the actual acquired or imported media.
Input Setup	Displays the input setup type used when the media was acquired.
Standard	Displays the PAL or NTSC video standard of the media.
ColorFX™	Displays any color effect applied to the clip.
TimeFX™	Displays the speed applied to the clip.
Audio Frequency	Displays the audio frequency of the clip.
Audio EQ	Displays the Audio EQ applied to the clip.
Rendered Media File Name	Displays the file name of the media that has been rendered.
Source Media File Size	Displays the file size of the source media in megabytes (MB) before any changes have been made to it during a Media 100 session.
Rendered Media File Size	Displays the file size of the rendered media in megabytes (MB) after rendering.
Audio In Channel	Displays the number of the Audio In Channel.
Audio Dynamics	Displays the Audio Dynamics applied to the clip

Bin Defaults Panel Elements *(Continued)*

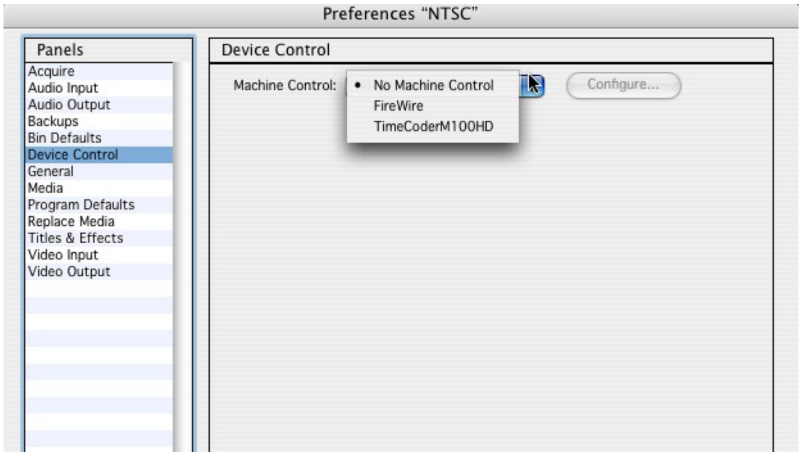
Name	Description
Reverb	Displays the Reverb applied to the clip.
Compressor	Displays the compressor used when the clip was acquired.
Frame Rate	Displays the frames per second of the acquired clip.
Interlacing	Displays the content as interlaced or progressive.
Bit Depth	Displays the bit depth of the acquired clip - 8-bit or 10-bit uncompressed.
Draft	Displays the content as draft or online quality.
Alpha	Displays the presence of an alpha channel on the acquired clip.
Input Scaling	Displays the type of input scaling

NOTE You can change the order that settings appear in the bin in List view by dragging the setting to a new position in the List View Columns section.

To change the view settings for a selected bin only, choose View>View Settings or select Settings in the Bin toolbar. You can change the order that settings appear in a specific bin in List View by dragging the column heading to the desired location.

Device Control Panel

The Device Control panel contains settings for VTR machine control.

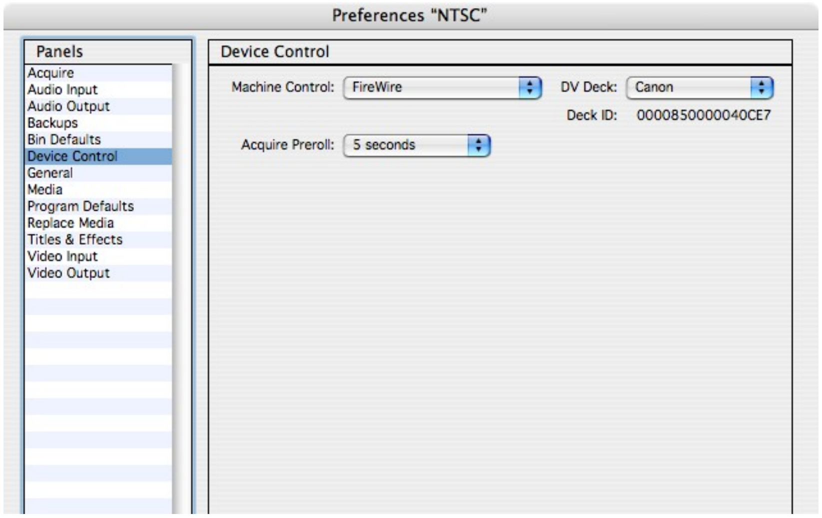


The following table describes the Device Control panel elements.

Device Control Panel Elements

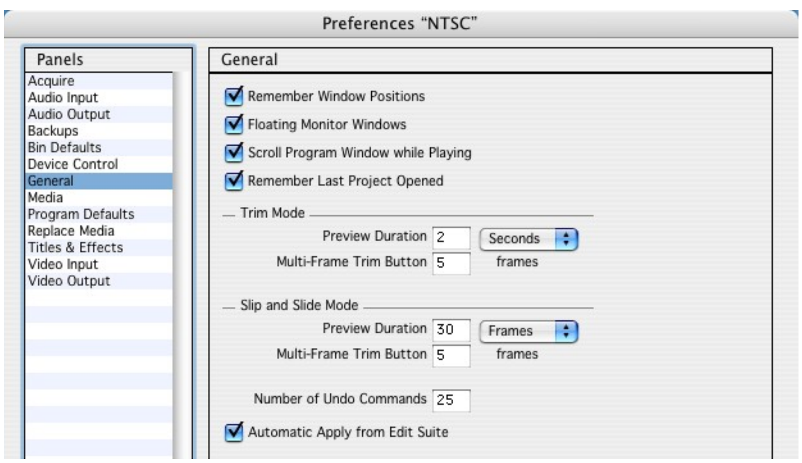
Name	Description
Machine control	Choose whether to use TimeCoderM100HD, FireWire or no machine control.
Configure	Click to open additional setting options for the TimeCoderM100HD.

When FireWire is the selected machine control, a selection for the type of DV Deck becomes available as well as a duration for preroll during acquisition.



General Panel

The General panel contains miscellaneous settings.



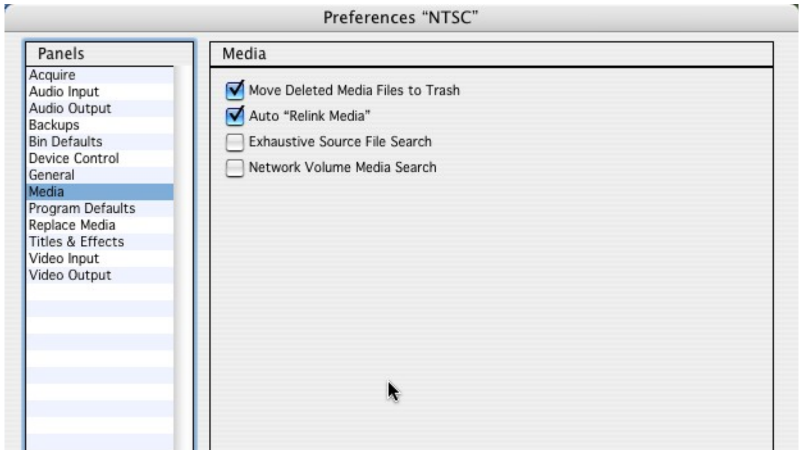
The following table describes the General panel elements.

General Panel Elements

Name	Description
Remember Window Positions check box	Remembers the position of your windows when you quit. When you launch the application, your windows appear in the same position as when you last quit the application.
Floating Monitor Window check box	Keeps the Monitor window in front of all other windows.
Scroll Program Window while Playing check box	Scrolls the Program window as you play a program to let you view each part of the program.
Remember Last Project Opened check box	Opens the last project saved the last time the application was open.
Trim Mode Preview Duration field and Multi-Frame Trim button	Sets the amount of time (seconds or frames) to preview a video selection at a trim point using Play Selected in the Edit Suite window. Half the time is allocated to preview the material before the trim point and half to preview it after the trim point. Multi-Frame Trim button specifies the number of frames to be trimmed when using the multi-frame trim button in the Edit Suite window.
Slip and Slide Mode Preview Duration field and Multi-Frame Trim Button field	Sets the amount of time (seconds or frames) to preview a video selection at a trim point using Play Selected in the Edit Suite window. Half the time is allocated to preview the material before the trim point and half to preview it after the trim point. Multi-Frame Trim button specifies the number of frames (from 2 to 30) that a trim point moves when you use Multi-Frame Forward and Multi-Frame Backward in the Edit Suite window.
Number of Undo Commands field	Specifies the number of commands that can be stored and available to undo (from 1 to 150). The default is 25. A higher number requires a greater amount of system memory.
Automatic Apply from Edit Suite	Applies changes made to a clip in the bin or timeline without clicking the Apply button.

Media Panel

The Media panel contains settings for manipulating media.



The following table describes the Media panel elements.

Media Panel Elements

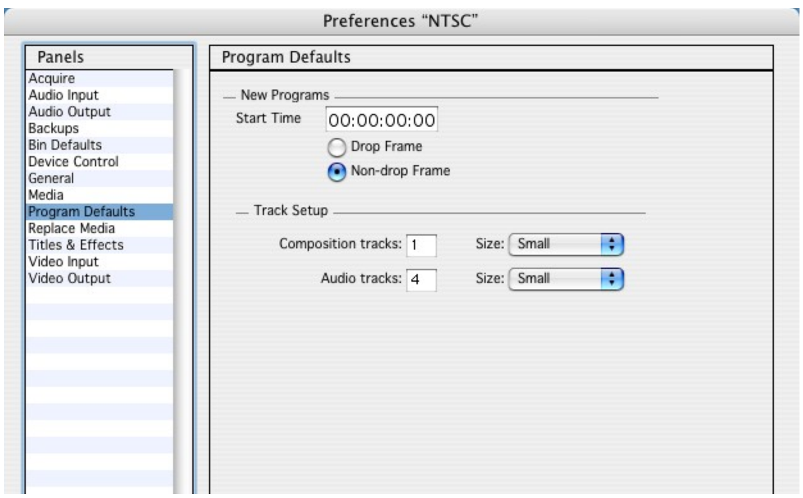
Name	Description
Move Deleted Media Files to Trash check box	Moves media files that you delete to the Trash. You can retrieve these files from the Trash or delete them by emptying the Trash. If you do not select this preference, the Media 100 program permanently removes deleted files from your hard drive.
Auto "Relink Media" check box	Relinks media in projects, bins, and programs to their source media files when you move media files to different folders or drives, change the names of folders or files, or open a bin owned by another project in a new project.

Media Panel Elements *(Continued)*

Name	Description
Exhaustive Source File Search check box	Searches the last place where source media was stored. If the source media is missing, it searches through all mounted drives and network volumes for PICT or QuickTime source files. Select this preference if you moved your source files and you intend to export, master to tape, render all, or force render all. If you do not select this preference, the Media 100 program only searches the last place the source file was found.
Network Volume Media Search check box	Searches in all network volumes for offline media when you open a project or select Relink Media.

Program Defaults Panel

The Program Defaults panel contains default settings for new programs.



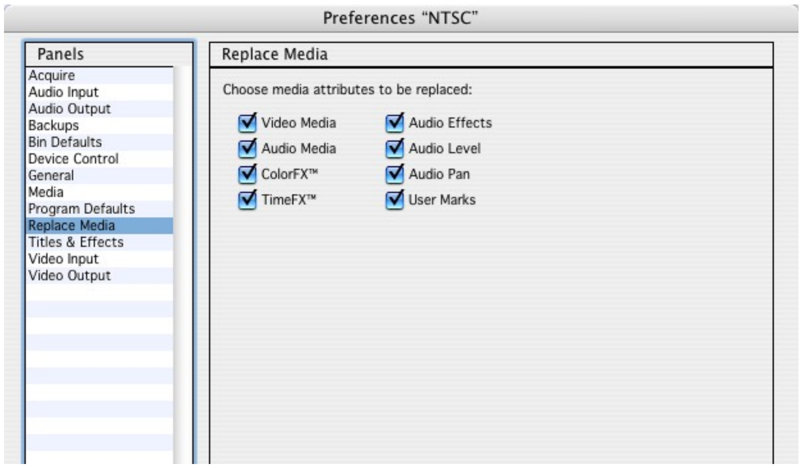
The following table describes the Program Defaults panel elements.

Program Defaults Panel Elements

Name	Description
Start Time text field	Set a specific start time for new programs.
Drop Frame radio button	Sets new NTSC programs to use drop frame timecode.
Non-drop Frame radio button	Sets new NTSC programs to use non-drop frame timecode.
Track Setup section	Set track setup defaults for the number of composition tracks and audio tracks for new programs. To change the track setup for the currently selected program, use the Track Setup dialog box (Track>Track Setup).
Audio Track Size	Set the expanded (Level and Pan display) audio track size for the entire project. Sizes include: ■ Small ■ Medium ■ Large To change the track size for the currently selected program, use the Track Setup dialog box (Track>Track Setup).
Video Track Size	Set the expanded (Opacity display) video track size for the entire project. Sizes include: ■ Small ■ Medium ■ Large To change the track size for the currently selected program, use the Track Setup dialog box (Track>Track Setup).

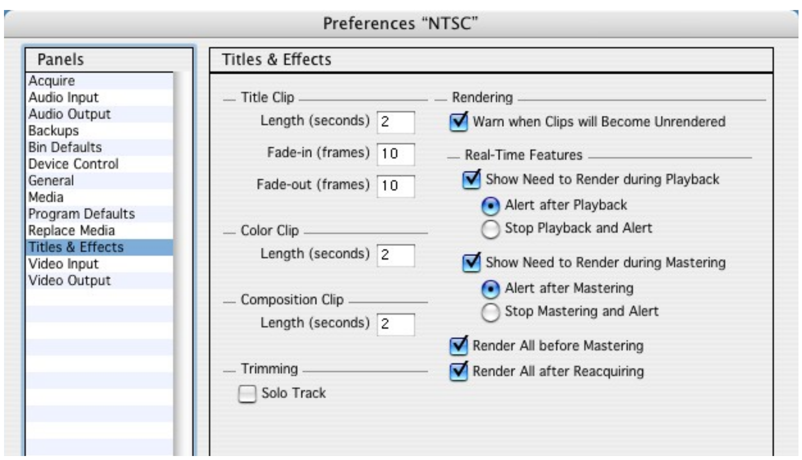
Replace Media Panel

The Replace Media panel lets you choose the clip attributes to replace when using the Replace Media command. See chapter 9 “Replacing Media.”



Titles & Effects Panel

The Titles & Effects panel contains settings for rendering titles and effects as well as title clip length, fade-in, and fade-out.



The following table describes the Titles & Effects panel elements.

Titles & Effects Panel Elements

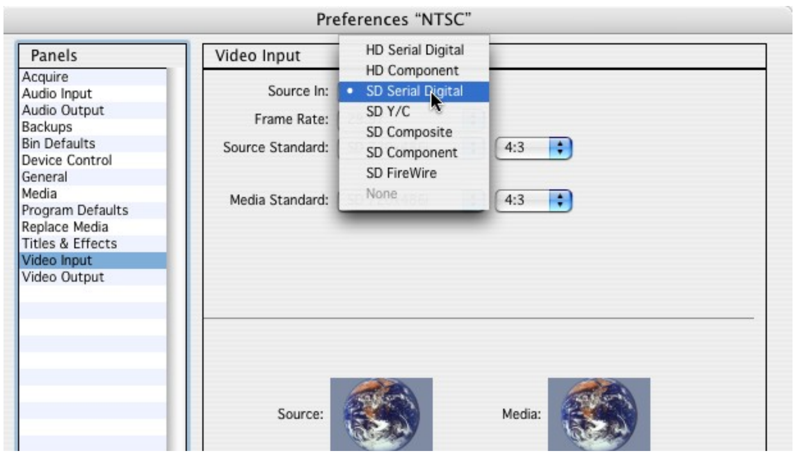
Name	Description
Length (seconds) field	Sets the default length of a title clip.
Fade-in (frames) field	Sets the default fade-in duration of a title clip.
Fade-out (frames) field	Sets the default fade-out duration of a title clip.
Color Clip Length (seconds) field	Sets the default length of a color clip.
Composition Clip Length (seconds) field	Sets the default length of a composition clip.
Render All before Mastering check box	Renders all titles and effects before mastering.
Render All after Reacquiring check box	Renders all titles and effects after reacquiring.
Warn when Clips will Become Unrendered check box	Alerts you if an action that you are performing on a clip, such as adding a transition or effect, causes the clip to unrender. Applies primarily to actions involving composition clips.
Show Need to Render During Playback check box	Notifies you if you need to render a title before you can preview it during playback. You can choose for the system to alert you when playback is complete or to stop playback and alert you.
Alert after Playback radio button	Alerts you that you need to render a title when playback is complete.
Stop Playback and Alert radio button	Stops playback and alerts you that you need to render a title before playback is complete.
Show Need to Render During Mastering check box	Notifies you if you need to render a title before mastering it to tape. You can choose for the system to alert you after mastering, or to stop mastering and alert you.

Titles & Effects Panel Elements *(Continued)*

Name	Description
Alert after Mastering radio button	Alerts you that you need to render a title when mastering is complete.
Stop Mastering and Alert radio button	Stops mastering and alerts you that you need to render a title before mastering is complete.

Video Input Panel

The Video Input panel contains settings for acquiring video media. This includes batch acquisition as well as conforming media from its source media standard to a media standard on disk.



The following table describes the Video Input panel elements.

Video Input Panel Elements

Name	Description
Source In	Specify the video source for your equipment.
Frame Rate	Specify the frames per second - 25 (PAL) or 29.97 (NTSC)

Video Input Panel Elements *(Continued)*

Name	Description
Source Standard	Specify the resolution; and aspect ratio **NOTE** Only HD media offers resolution selections.
Media Standard	Specify the format to convert the video source to.
Video Pedestal	Specify the playback reference black levels for the NTSC standard. ■ 0 IRE - The reference black signal for NTSC in Japan ■ 7.5 IRE - The reference black signal for NTSC in countries other than Japan **NOTE** The Video Pedestal selection only displays if the Video Input Standard is NTSC Composite, Y/C or Component analog. The pedestal is always 0 IRE for PAL video standards
Convert	Specify how to convert the video source. Options include include 4:3 to 14:9, 16:9 to 14:9, Crop, Pillarbox and Scale to Fit. In addition a Custom Scaler offers complete flexibility of conversion options The Convert button only displays if the Video Input Standard differs from the Media Standard.

Conversion Descriptions

Convert	Description
4:3 to 14:9	Conversion to BBC recommended standard. Crops the top and bottom, and pillarboxes the result (black on left and right)
16:9 to 14:9	Conversion to BBC recommended standard. Crops the left and right, and letterboxes the result (black on top and bottom)
Crop	Crops the top and bottom.
Pillarbox	Adds black on left and right.
Scale to Fit	Stretches the image to fit the new aspect ratio.
Crop to 4:3	Crops the left and right.
Letterbox	Adds black on top and bottom.

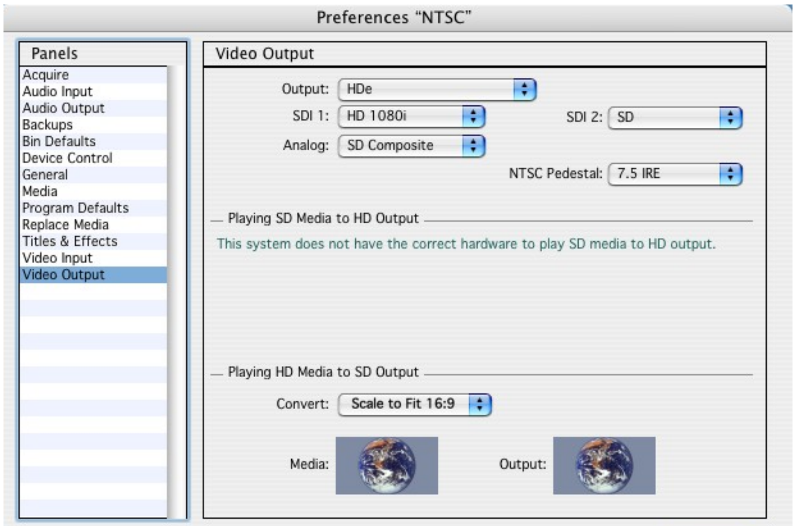
NOTE Media 100 systems allow for one conversion. Therefore, if you convert the source content to a different standard during acquisition or import, that is the standard the content must be output to. However, if the source content is acquired in it's native form it can be converted on output. Again, only one conversion is allowed. (Media 100 HDx enables multiple conversions.) Also note, the conversion options that are available are dependent upon the Media 100 system being used.

NOTE The Video Input Preference Panel does not allow SD media to be converted from a 4:3 aspect ratio to a 16:9 aspect ratio or vice versa. To accomplish this conversion, acquire the content at it's default aspect ratio, then use the Conform dialog box to convert to the desired aspect ratio.

NOTE If you choose 720p input your external monitors will display garbage. Go to the Video Output preference panel and set SDI 1 to 720 to see your video. However, this will blank out any external monitor using SDI SD.

Video Output Panel

The Video Output panel contains settings for the output of video media. This includes mastering to tape and conforming media for output.



The following table describes the Video Output panel elements.

Video Output Panel Elements

Name	Description
Output	Specifies the Media100 system; FireWire; FireWire (Mastering Only) or None for video output.
SDI 1	Specify output options - HD 1080i, HD 720p, SD
SDI 2	not currently active (reserved for future use)
Analog	Specify output options - SD Composite, SD Component, HD Component, HD VGA
NTSC Pedestal	Specify the playback reference black levels for the NTSC standard. ■ 0 IRE - The reference black signal for NTSC in Japan ■ 7.5 IRE - The reference black signal for NTSC in countries other than Japan **NOTE** The Video Pedestal selection only displays if the Video Input Standard is NTSC Composite, Y/C or Component analog. The pedestal is always 0 IRE for PAL video standards
Playing 4:3 SD Media to HD Output Convert	Specify output options - 4:3 to 14:9; 16:9 to 14:9; Crop; Pillarbox; Scale to Fit
Playing 4:3 SD Media to HD Output Custom button	Activates Custom Scaler dialog box for Crop, Side Panel, Tilt and Anamorphism adjustments.
Playing 4:3 SD Media to HD Output Save button	Saves the custom conversion setting to the selection list.
Playing 4:3 SD Media to HD Output Delete button	Deletes the custom conversion setting from the selection list.

Video Output Panel Elements *(Continued)*

Name	Description
Playing HD Media to SD Output Convert	Specify output options - 4:3 to 14:9; 16:9 to 14:9; Crop to 4:3; Letterbox to 4:3; Scale to Fit
Playing HD Media to SD Output Custom button	Activates Custom Scaler dialog box for Crop, Side Panel, Tilt and Anamorphism adjustments.
Playing HD Media to SD Output Save button	Saves the custom conversion setting to the selection list.
Playing HD Media to SD Output Delete button	Deletes the custom conversion setting from the selection list.

NOTE Only one conversion on a source clip is permitted. Therefore, if source media is converted upon input it can not be converted to something other on output.

NOTE If FireWire is the output selection and a deck is not connected, the timeline will not play because it’s attempting to master the timeline as output and it can’t. Therefore, if FireWire (Mastering Only) is the selected output, the system will allow the timeline to be played and will seek the deck connection when mastering only.

▼ Quick Launch of Third-Party Applications

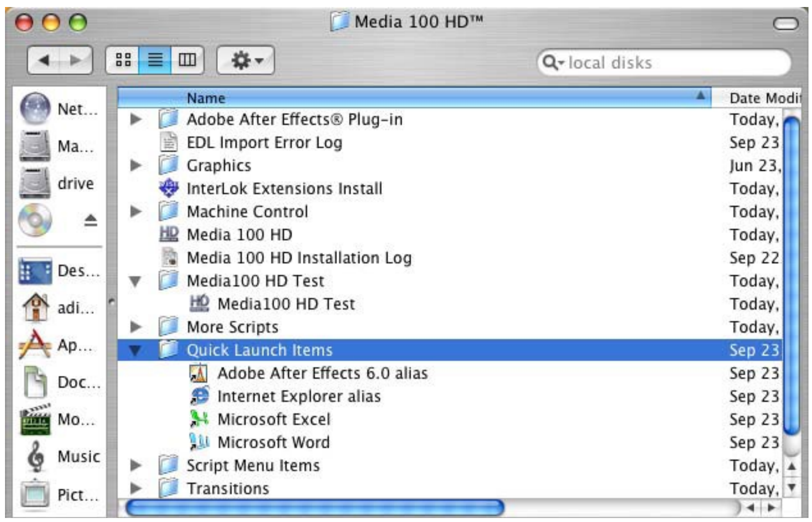
Quick Launch lets you open third-party applications from within Media 100.

NOTE You can use Quick Launch to access aliases of any kind, including documents, folders, scripts, servers, and so on.

To set up the Quick Launch feature

1 Create aliases of the items you want to appear in the Quick Launch submenu.

2 Move the aliases to the Quick Launch Items folder in the Media 100 folder.



To use Quick Launch

➤ Choose Media 100>Launch and select the alias to launch from the submenu.

The Finder switches control to the selected application. Media 100 runs in the background; you can switch to it as necessary.

Although you can put an unlimited number of aliases in the Quick Launch Items folder, only the first 30 appear in the Quick Launch submenu. The submenu is updated each time you switch from another application to Media 100, making it possible to switch to the Finder, modify the Quick Launch Items folder, and then switch back to Media 100 to use the new items.

NOTE Although you can put an application into the Quick Launch Items folder, it is not recommended. An alias is more efficient and keeps the folder size smaller.